

This skills check covers pre-requisite material needed for the second unit of the course, derivatives. You should practice these ideas to ensure that you can quickly and correct answer such questions. Being able to do so will make learning the *calculus* material of the third unit much easier. It's never fun to be held up by Algebra/Trig. when trying to learn Calculus! *This is just a sample for practice. You may receive help on these questions and work with other students.*

1. Compositions of functions.

- (a) Express the function $y = 8(2x+8)^2$ as a composition of functions $f(g(x))$.

Many answers are possible.
 $f(x) = 8x^2$ and $g(x) = 2x+8$

- (b) Express the function $y = \cos(3x)$ as a composition of functions $f(g(x))$.

Many answers are possible.
 $f(x) = \cos(x)$ and $g(x) = 3x$

- (c) Express the function $y = e^{\sqrt{x+6}} + 7$ as a composition of functions $f(g(x))$.

Many answers are possible.
 $f(x) = e^x + 7$ and $g(x) = \sqrt{x+6}$

2. Solving equations for a given variable.

- (a) Solve the following equation for z : $3y^2z - 8yz = 5x^4 + 12x^3$.

$$3y^2z - 8yz = 5x^4 + 12x^3 \Leftrightarrow z(3y^2 - 8y) = 5x^4 + 12x^3 \Leftrightarrow z = \frac{5x^4 + 12x^3}{3y^2 - 8y}$$

- (b) Solve the following equation for z : $2yz \cos(y) + 2x \sin(x) = -5z \sin(y^2) + \cos(x^2)$.

$$\begin{aligned} 2yz \cos(y) + 2x \sin(x) &= -5z \sin(y^2) + \cos(x^2) \\ \Leftrightarrow 2yz \cos(y) + 5z \sin(y^2) &= \cos(x^2) - 2x \sin(x) \\ \Leftrightarrow z[2y \cos(y) + 5 \sin(y^2)] &= \cos(x^2) - 2x \sin(x) \\ \Leftrightarrow z &= \frac{\cos(x^2) - 2x \sin(x)}{2y \cos(y) + 5 \sin(y^2)} \end{aligned}$$

- (c) Solve the following equation for b_1 : $A = \frac{1}{2}h(b_1 + b_2)$

$$A = \frac{1}{2}h(b_1 + b_2) \Leftrightarrow \frac{2A}{h} = b_1 + b_2 \Leftrightarrow b_1 = \frac{2A}{h} - b_2$$

3. Equations of lines.

- (a) Find the equation of the line through the points $(1, -2)$ and $(1, 7)$.

$$m = \frac{7 - (-2)}{1 - 1} = \frac{9}{0} \text{ undefined} \quad \text{Therefore we have a vertical line.}$$

The line is $x = 1$.

- (b) Find the equation of the line that passes through $(1, 5)$ and is parallel to $y - 5x = 1$.

$$y - 5x = 1 \Leftrightarrow y = 5x + 1$$

Our line must have slope 5.

$$\text{Therefore our line is: } y - 5 = 5(x - 1) \Leftrightarrow y = 5x$$

- (c) Find an equation of the line that passes through $(5, 2)$ and is perpendicular to $4x + 3y = 12$.

$$4x + 3y = 12 \Leftrightarrow y = -\frac{4}{3}x + 3$$

Our line must have slope $\frac{3}{4}$.

$$\text{Therefore our line is: } y - 2 = \frac{3}{4}(x - 5) \Leftrightarrow y = \frac{3}{4}x - \frac{7}{4}$$

4. Properties of Logarithms.

- (a) Expand the following using properties of logarithms: $\ln \left(\frac{x^2 \sqrt{1-x^2}}{(x^3+1)^2} \right)$

$$\ln \left(\frac{x^2 \sqrt{1-x^2}}{(x^3+1)^2} \right) \Leftrightarrow \ln(x^2) + \ln(\sqrt{1-x^2}) - \ln((x^3+1)^2) \Leftrightarrow 2\ln(x) + \frac{1}{2}\ln(1-x^2) - 2\ln(x^3+1)$$

- (b) Expand the following using properties of logarithms: $\ln \left(\sqrt[4]{\frac{x^2(x+1)}{2-x}} \right)$

$$\begin{aligned} \ln \left(\sqrt[4]{\frac{x^2(x+1)}{2-x}} \right) &\Leftrightarrow \frac{1}{4} \ln \left(\frac{x^2(x+1)}{2-x} \right) \Leftrightarrow \frac{1}{4} [\ln(x^2) + \ln(x+1) - \ln(2-x)] \\ &\Leftrightarrow \frac{1}{4} [2\ln(x) + \ln(x+1) - \ln(2-x)] \end{aligned}$$

(c) Expand the following using properties of logarithms: $\ln\left(\frac{2x-3}{x^3\sqrt{x+4}}\right)$

$$1\ln\left(\frac{2x-3}{x^3\sqrt{x+4}}\right) \Leftrightarrow \ln(2x-3) - \ln(x^3) - \ln(\sqrt{x+4}) \Leftrightarrow \ln(2x-3) - 3\ln(x) - \frac{1}{2}\ln(x+4)$$

5. Trig values and Trig ratios.

(a) Compute the value of the each of the following:

$$\begin{array}{lll} - \cos\left(\frac{7\pi}{6}\right) = -\frac{\sqrt{3}}{2} & - \sin\left(\frac{3\pi}{2}\right) = -1 & - \tan\left(\frac{3\pi}{4}\right) = -1 \\ - \sec\left(\frac{4\pi}{3}\right) = -2 & - \csc\left(\frac{\pi}{3}\right) = \frac{2}{\sqrt{3}} & - \cot\left(\frac{5\pi}{6}\right) = -\sqrt{3} \end{array}$$

(b) Set up a trig. ratio to relate the variables in the triangle below (on the left).

$$\sin(\theta) = \frac{1/2}{5} \Rightarrow \sin(\theta) = \frac{5}{2}$$

(c) Set up a trig. ratio to relate the variables in the diagram below (on the right).

$$\tan(\theta) = \frac{10}{3}$$

